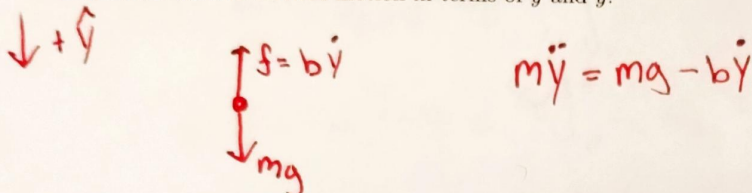


1. A particle of mass m is released from rest at $t = 0$. Take the positive y -direction to be vertically downward. The particle experiences gravity and a linear drag force

$$\mathbf{f} = -b\dot{\mathbf{r}},$$

where b is a positive constant.

- (a.) Draw and label a free-body diagram for the particle after it is released. Use the diagram to write Newton's second law for the vertical motion in terms of $\dot{y}$ and $\ddot{y}$.



- (b.) Find the terminal velocity $\dot{y}_{\text{ter}}$. Briefly state the condition that defines terminal velocity.

$$m\ddot{y} = 0 \quad mg = b\dot{y} \quad , \quad \dot{y}_{\text{ter}} = \frac{mg}{b}$$

$$\dot{y}_{\text{ter}} = \tau g$$

- (c.) Define the time constant

$$\tau = \frac{m}{b}.$$

Using $\dot{y}_{\text{ter}}$ and τ , separate and integrate the equation of motion to find $\dot{y}(t)$. Your answer must satisfy

$$\dot{y}(0) = 0.$$

$$m\ddot{y} = mg - b\dot{y} \xrightarrow{\text{divide } m} \ddot{y} = g - \frac{b}{m}\dot{y} \quad , \quad g = \frac{\dot{y}_{\text{ter}}}{\tau} \quad \frac{b}{m} = \frac{1}{\tau}$$

$$\ddot{y} = \frac{\dot{y}_{\text{ter}} - \dot{y}}{\tau} \rightarrow \frac{d\dot{y}}{dt} = \frac{\dot{y}_{\text{ter}} - \dot{y}}{\tau}$$

$$\int_0^{\dot{y}(t)} \frac{d\dot{y}}{\dot{y}_{\text{ter}} - \dot{y}} = \int_0^t \frac{1}{\tau} dt \rightarrow -\ln\left(\frac{\dot{y}_{\text{ter}} - \dot{y}(t)}{\dot{y}_{\text{ter}}}\right) = \frac{t}{\tau}$$

$$\frac{\dot{y}_{\text{ter}} - \dot{y}(t)}{\dot{y}_{\text{ter}}} = e^{-t/\tau} \rightarrow \therefore \dot{y}(t) = \dot{y}_{\text{ter}} (1 - e^{-t/\tau})$$

or

$$\dot{y}(t) = \frac{mg}{b} (1 - e^{-bt/m})$$

2. A particle has an initial horizontal velocity $\dot{x}(0) = v_{0x}$ and experiences a linear drag force. Its horizontal velocity is

$$\dot{x}(t) = v_{0x} e^{-t/\tau},$$

where $\tau = m/b$.

- (a.) Integrate $\dot{x}(t)$ to find the horizontal position $x(t)$. Use the initial condition $x(0) = 0$.

$$\dot{x}(t) = \frac{dx}{dt} = v_{0x} e^{-t/\tau}, \quad \int_0^{x(t)} dx = \int_0^t v_{0x} e^{-t/\tau} dt$$

$$x(t) = v_{0x} [-\tau e^{-t/\tau}]_0^t = v_{0x} \tau (1 - e^{-t/\tau})$$

$$\therefore x(t) = v_{0x} \tau (1 - e^{-t/\tau})$$

- (b.) Find the limiting horizontal position $x(\infty)$. Briefly explain the physical meaning of this result.

$$t \rightarrow \infty$$

$$x(\infty) = v_{0x} \tau (1 - \cancel{e^{-\infty/\tau}}) = v_{0x} \tau$$

$$\therefore x(\infty) = v_{0x} \tau = \frac{m v_{0x}}{b}$$

The particle can only travel a finite distance as the horizontal drag reduces its velocity to zero as $t \rightarrow \infty$